Exploratory Data Analysis on a Cars Dataset

Clean, explore, visualise, and explain meaningful patterns in automobile data

Project Title	Exploratory Data Analysis on a Cars Dataset
Student Name	P Rehan Khan
USN / Batch	1BF25AI099
Institution	BMS College of Engineering
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Dataset Source	Vehicle dataset from CarDekho Dataset link (click to open)

2. Objective

The goal of this project is to perform a complete Exploratory Data Analysis (EDA) on a used-car dataset to understand automobile pricing patterns, identify outliers, and find relationships between mileage, vehicle age, fuel type and price. No machine-learning model is built; the focus is on data quality, visual exploration and evidence-based insights.

Questions selected for analysis:

- How are car prices distributed?
- Which car brands are the most common?
- Does mileage (km driven) appear to influence price?
- Do automatic and manual cars differ in price?
- Which numerical features are strongly correlated, and how does vehicle age relate to price?
- How does the average price differ across fuel types?

3. Dataset Overview

The dataset (Vehicle dataset from CarDekho) contains **180 rows and 20 columns** after cleaning (the same size as before cleaning, since no rows needed removal). The **target variable** is **price_usd**, the selling price of each car, and the analysis studies how the other attributes relate to it.

Numerical and categorical variables

Type	Count	Columns
Numerical	11	car_id, year, engine_cc, power_hp, fuel_economy_mpg, seats, km_driven, num_owners, accident_history, price_usd, car_age
Categorical	9	brand, model, segment, body_type, fuel_type, transmission, condition, color, city

Feature descriptions and data types

Column	Dtype	Description
car_id	int64	Unique identifier of the listing
brand / model	object	Manufacturer and model name (e.g. Hyundai i20)
year / car_age	int64	Manufacturing year; age in years (2026 minus year)
segment / body_type	object	Market segment (Hatchback, SUV, MPV...) and body style
fuel_type	object	Petrol, Diesel, CNG, Hybrid or Electric
transmission	object	Manual or Automatic
engine_cc / power_hp	int64	Engine displacement (cc) and power output (hp)
fuel_economy_mpg	float64	Fuel economy in miles per gallon
seats	int64	Number of seats
km_driven / num_owners	int64	Kilometres driven; number of previous owners
condition	object	Condition grade (e.g. Good, Fair, Excellent)
accident_history	int64	Accident indicator stored as an integer code
color / city	object	Car colour and city of listing
price_usd	float64	Selling price in USD (target variable)

Sample records (first 5 rows, selected columns)

brand	model	year	fuel	trans.	engine_cc	hp	km_driven	price_usd
Hyundai	i20	2012	Diesel	Manual	1320	78	120,639	1,460
Volkswagen	Polo	2021	Petrol	Automatic	1000	116	39,084	7,710
Volkswagen	Golf	2020	Petrol	Automatic	1400	136	39,194	11,100
Tata	Nexon	2019	Diesel	Manual	1320	115	69,792	4,240
Mahindra	Scorpio	2024	Diesel	Manual	2420	148	9,464	17,960

4. Data Quality and Cleaning

The following checks and decisions were applied in the cleaning step of the notebook:

Check / Step	Result and decision
Shape, columns, dtypes	180 rows x 20 columns; 11 numerical and 9 categorical columns. All data types were already suitable, so no type conversion was required.
Missing values	**0 missing values** in every column (180 non-null entries each). Median (numerical) / mode (categorical) imputation was coded as a safeguard, but nothing needed to be imputed, so data integrity is fully preserved.
Duplicate rows	**0 duplicates** were found; the drop-duplicates step therefore removed nothing.
Column names / labels	Column names were standardised (stripped and lower-cased) and the brand label was reduced to its first word, so categories stay consistent for grouping.
Derived feature	**car_age** was recalculated as 2026 minus year so that age is consistent with the current year.
Outliers	High-priced cars (about 40,000 USD and above) and very high-mileage cars (above about 250,000 km) are visible on the charts. They were **kept**: they look like genuine premium or heavily-used vehicles rather than errors, and removing them would hide real price variation.

Dataset size before and after cleaning

Stage	Rows	Columns	Missing values	Duplicates
Before cleaning	180	20	0	0
After cleaning	180	20	0	-

5. Descriptive Analysis

Key numerical ranges, read from the notebook's charts (values are approximate):

Variable	Approximate range / typical value	Source
price_usd	Minimum about 1,000; maximum about 51,000; most cars below 10,000 (right-skewed)	Chart 1
km_driven	From under 10,000 up to about 263,000 km	Chart 3
Manual median price	About 4,700 USD (box ends near 2,800 and 10,000)	Chart 4
Automatic median price	About 8,100 USD (box ends near 3,400 and 17,200)	Chart 4
Mean price by fuel type	Hybrid about 10,600; Petrol about 10,400; Diesel about 10,400; CNG about 7,900; Electric about 5,700	Chart 6

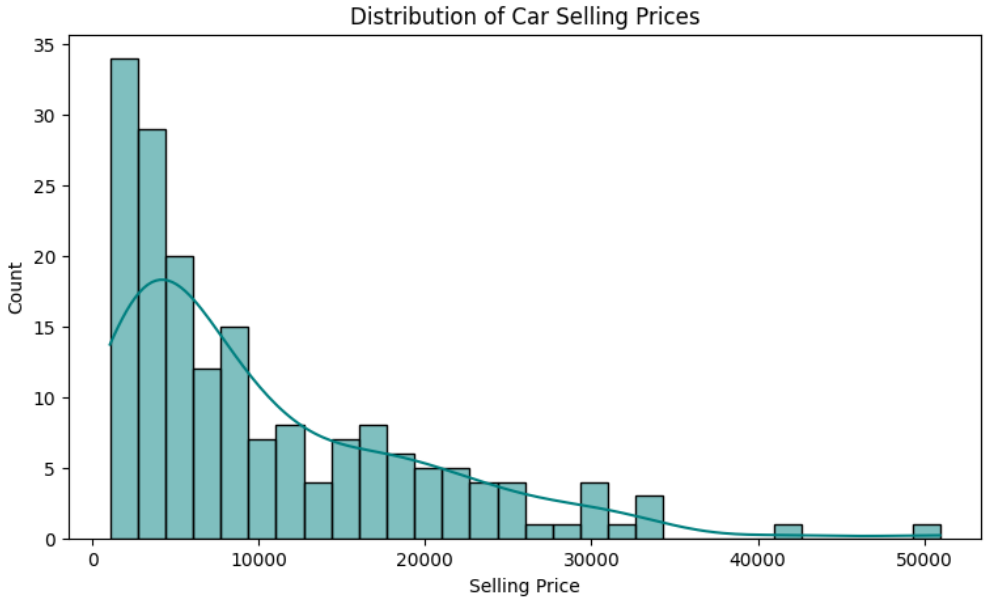
Categorical summary: the most common brands are Volkswagen (19 cars), Maruti (18), Chevrolet (16), Ford (15), and Hyundai and Audi (14 each). Petrol is the most common fuel type; Electric, Hybrid and CNG cars appear only a few times each (Chart 3).

Observation: the average price is pulled above the typical price by a small number of expensive cars, so the **median is a better measure** of a typical car than the mean in this dataset.

6. Visual EDA

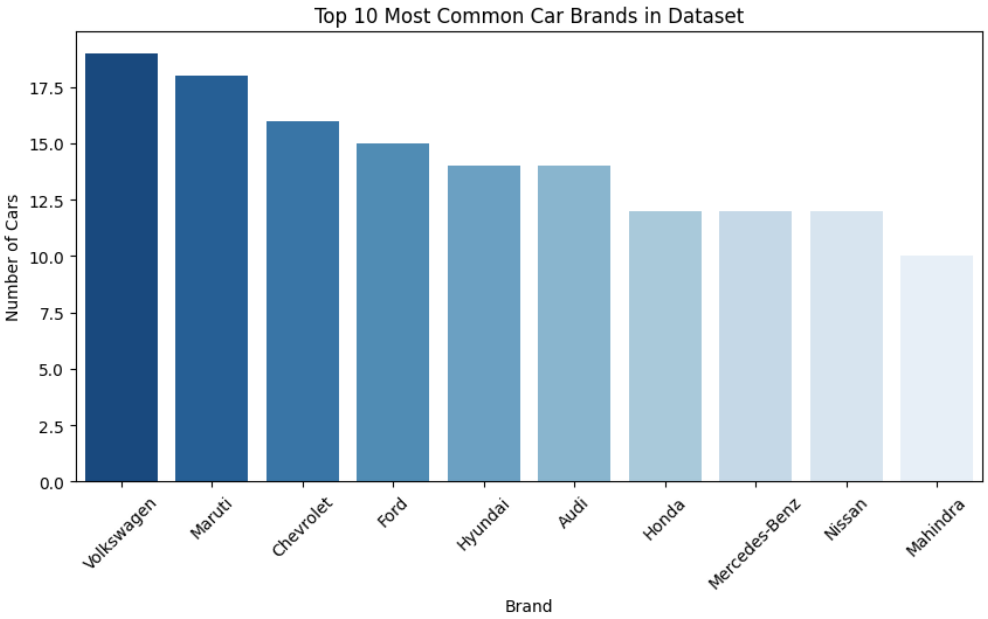
Six charts of four types (histogram, bar chart, scatter plot, box plot, heatmap) are shown below, each with an observation.

Chart 1: Distribution of Car Selling Prices (Histogram)



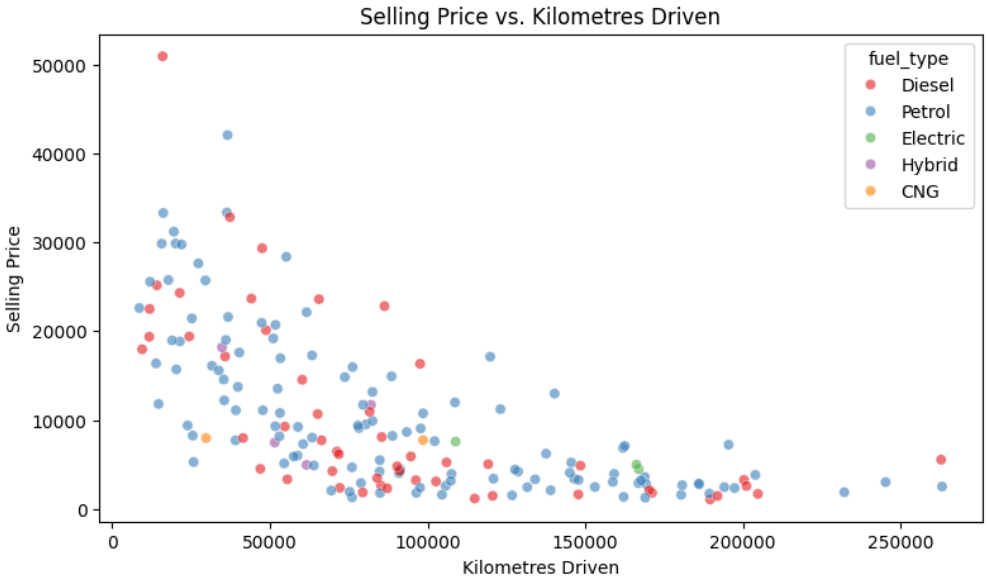
Observation: Prices are strongly right-skewed. The tallest bars are at the low end (about 1,000 to 6,000 USD), with roughly 34 and 29 cars in the first two bins. The count falls steadily as price rises, and only a couple of cars reach about 42,000 to 51,000 USD.

Chart 2: Top 10 Most Common Car Brands (Bar Plot)



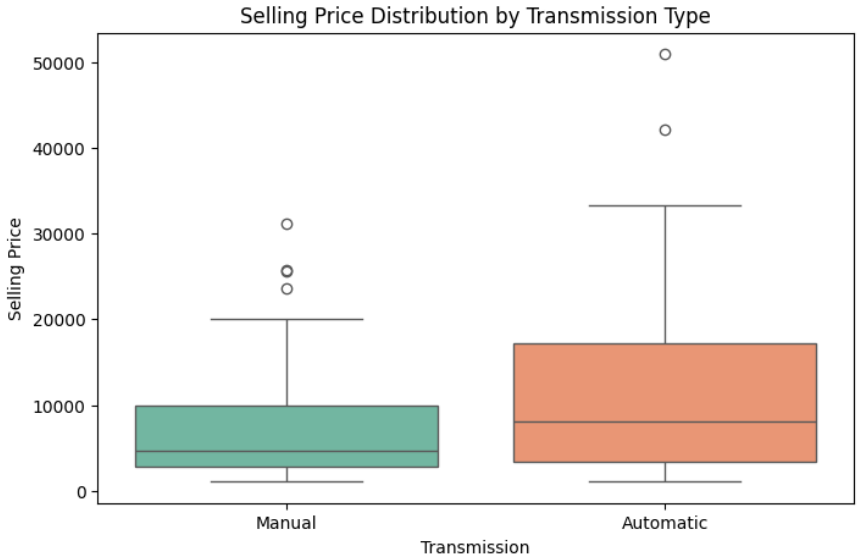
Observation: Volkswagen (19) and Maruti (18) lead, followed by Chevrolet (16) and Ford (15). The gap between 1st and 10th place (Mahindra, 10) is small, so no single brand dominates the dataset.

Chart 3: Selling Price vs. Kilometres Driven by Fuel Type (Scatter Plot)



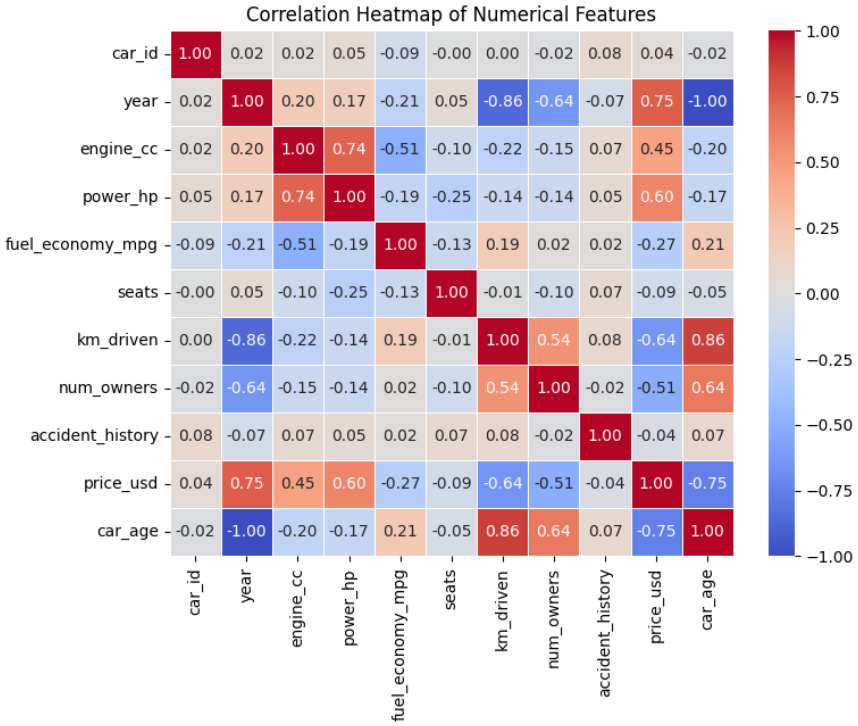
Observation: Price falls as kilometres driven rise: cars under 50,000 km span the whole price range, while cars above 150,000 km are almost all below 8,000 USD. Petrol and Diesel dominate the points; Electric, Hybrid and CNG have too few cars to compare reliably.

Chart 4: Selling Price by Transmission Type (Box Plot)



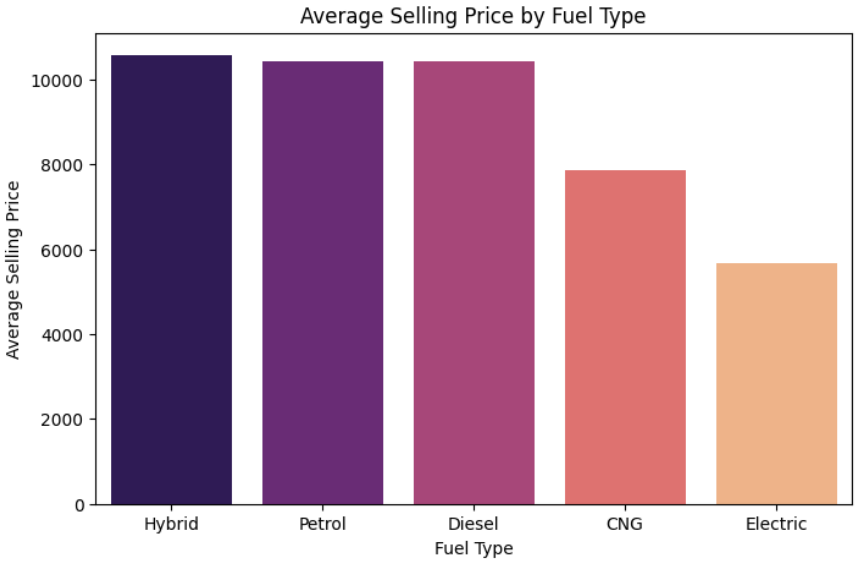
Observation: Automatic cars have a higher median price (about 8,100 USD) than manual cars (about 4,700 USD) and a much wider spread. Both groups show high-price outliers, the most extreme being automatic cars at about 42,000 and 51,000 USD.

Chart 5: Correlation Heatmap of Numerical Features (Heatmap)



Observation: Price correlates positively with year (0.75) and power_hp (0.60), and negatively with car_age (-0.75), km_driven (-0.64) and num_owners (-0.51). km_driven is strongly linked to age (0.86), so mileage and age partly carry the same information. Engine size and power are also related (0.74). car_age and year are perfectly inverse (-1.00) because one is derived from the other.

Chart 6: Average Selling Price by Fuel Type (Bar Plot)



Observation: Hybrid, Petrol and Diesel have nearly the same average price (about 10,400 to 10,600 USD). CNG averages about 7,900 USD and Electric about 5,700 USD. Because Electric, Hybrid and CNG have very few cars, those averages are less reliable.

7. Key Findings

1. **Prices are right-skewed.** Most cars sell for under about 10,000 USD, while a few premium cars reach 42,000 to 51,000 USD (Chart 1).
2. **Age is the strongest driver of price.** Price has a correlation of +0.75 with year and -0.75 with car age, so newer cars are worth clearly more (Chart 5).
3. **Higher mileage goes with lower prices.** km_driven has a correlation of -0.64 with price, and high-km cars cluster at low prices (Charts 3 and 5). Mileage and age are closely linked (0.86), so their effects cannot be fully separated here.
4. **Automatic cars carry a price premium.** Their median price (about 8,100 USD) is roughly 1.7 times that of manual cars (about 4,700 USD) (Chart 4). This is an association; automatic cars may also be newer or more powerful.
5. **Power matters more than engine size for price.** power_hp (0.60) correlates more strongly with price than engine_cc (0.45) (Chart 5).
6. **Fuel type shows a modest price difference.** Petrol, Diesel and Hybrid cars average about 10,400 to 10,600 USD, with CNG and Electric lower (Chart 6). Diesel cars do not show a higher average than petrol cars in this data.
7. **The brand mix is balanced.** Volkswagen, Maruti, Chevrolet and Ford are the most frequent brands, with no single brand dominating (Chart 2).

Limitations of the dataset

- Only **180 records**, so results for individual brands and models are based on small counts.
- Electric, Hybrid and CNG cars appear only a few times, so fuel-type averages for them are unreliable.
- No information on **service history**, tyre or battery condition, or market demand, all of which strongly affect real-world prices.
- Prices are given in USD while the listing cities are Indian cities, so the currency basis should be verified before drawing market conclusions.
- The analysis shows **associations, not causes** (for example, transmission and price).

8. Conclusion

This EDA examined a 180-record used-car dataset with 20 attributes. The data was complete, with no missing values and no duplicates, so cleaning focused on standardising column names and labels, confirming data types and deciding to keep the genuine high-price and high-mileage outliers. The analysis shows that vehicle age and mileage are the main factors lowering price, that automatic and more powerful cars sell for more, and that fuel type has a smaller effect.

Challenges faced: standardising column names so that the code ran without KeyError problems, handling a right-skewed price variable, and recognising that age and mileage are highly correlated and therefore hard to separate.

Future analysis: compare prices within a single segment or body type to remove mix effects, analyse brand-wise depreciation, test the transmission premium after controlling for age, use a larger dataset with more Electric and Hybrid cars, and build a machine-learning model (for example regression) to predict selling price.